Based on the latest CBSE blueprint and a rigorous analysis of Previous Year Question (PYQ) trends, here is a highly realistic, predicted 80-mark Class 12 Maths board paper for the 2027 examination [1-4].

Calculus dominates with 35 marks, followed by Vectors & 3D Geometry (14 marks), Algebra (10 marks), Probability (8 marks), Relations & Functions (8 marks), and Linear Programming (5 marks) [5]. The paper strictly follows the official 38-question structure.

PREDICTED CBSE CLASS 12 MATHS BOARD PAPER 2027

SECTION A: Objective Type Questions (1 Mark Each x 20 = 20 Marks)

This section focuses on property-based formulas and quick conceptual applications [6, 7].

Q1. Let R be the relation in the set $A = \{1, 2, 3\}$ given by $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3)\}$. The relation R is reflexive but neither symmetric nor transitive. (True/False or MCQ format) * **Chapter:** Relations and Functions * **Probability Tag: Very High**

Q2. Find the principal value of $\cos^{-1}(\cos \frac{7\pi}{6})$. [8] * **Chapter:** Inverse Trigonometric Functions * **Probability Tag: Very High**

Q3. If A is a square matrix of order 3 such that $|A| = 5$, find the value of $|adjA|$. * **Chapter:** Determinants * **Probability Tag: Very High**

Q4. If A is a skew-symmetric matrix of an odd order, then what is the value of its determinant $|A|$? * **Chapter:** Matrices * **Probability Tag: High**

Q5. What is the derivative of $\sin^{-1}(2x\sqrt{1 - x^2})$ with respect to x? * **Chapter:** Continuity and Differentiability * **Probability Tag: High**

Q6. The function $f(x) = x^3 - 3x^2 + 3x - 100$ is strictly increasing in which interval? * **Chapter:** Application of Derivatives * **Probability Tag: Medium**

Q7. Evaluate: $\int e^x(\sin x + \cos x)dx$. [9] * **Chapter:** Integrals * **Probability Tag: Very High**

Q8. Determine the order and degree (if defined) of the differential equation: $\left(\frac{d^2y}{dx^2}\right)^2 + \cos\left(\frac{dy}{dx}\right) = 0$. [10] * **Chapter:** Differential Equations * **Probability Tag: Very High**

Q9. Find the integrating factor of the linear differential equation: $\frac{dy}{dx} + \frac{y}{x} = x^2$. [11] * **Chapter:** Differential Equations * **Probability Tag: High**

Q10. Find the projection of the vector $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ on the vector $\vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$. [12] * **Chapter:** Vector Algebra * **Probability Tag: Very High**

Q11. Find the area of the parallelogram whose adjacent sides are determined by the vectors $\vec{a} = \hat{i} - \hat{j} + 3\hat{k}$ and $\vec{b} = 2\hat{i} - 7\hat{j} + \hat{k}$. * **Chapter:** Vector Algebra * **Probability Tag: High**

Q12. Find the direction cosines of a line which makes equal angles with the coordinate axes. [13, 14] * **Chapter:** Three-Dimensional Geometry * **Probability Tag: Very High**

Q13. Find the angle between the lines whose direction ratios are 3, 2, -6 and 1, 2, 2. [13] * **Chapter:** Three-Dimensional Geometry * **Probability Tag: Medium**

Q14. In a Linear Programming Problem, if the feasible region is bounded and the corner points are (0,0), (5,0), (3,4), and (0,5), find the maximum value of the objective function $Z = 3x + 2y$. * **Chapter:** Linear Programming * **Probability Tag: Very High**

Q15. The probability that A hits a target is 1/3 and the probability that B hits it is 2/5. If they shoot independently, what is the probability that the target will be hit? [15] * **Chapter:** Probability * **Probability Tag: High**

Q16. Find the value of constant k so that the function $f(x) = kx + 1$ for $x \leq \pi$ and $f(x) = \cos x$ for $x > \pi$ is continuous at $x = \pi$. [16] * **Chapter:** Continuity and Differentiability * **Probability Tag: Very High**

Q17. Evaluate the definite integral using properties: $\int_{-\pi/2}^{\pi/2} \sin^7 x \, dx$. * **Chapter:** Integrals * **Probability Tag: High**

Q18. A fair die is tossed twice. What is the probability of getting a 4, 5, or 6 on the first toss and a 1, 2, 3, or 4 on the second toss? [15] * **Chapter:** Probability * **Probability Tag: Medium**

Q19. (Assertion-Reason): Assertion (A): The domain of the function $y = \sin^{-1} x$ is $[-1, 1]$. **Reason (R):** The principal value branch of $\sin^{-1} x$ has the range $[-\pi/2, \pi/2]$. * **Chapter:** Inverse Trigonometric Functions * **Probability Tag: High**

Q20. (Assertion-Reason): Assertion (A): Two vectors $\vec{a}$ and $\vec{b}$ are collinear if their direction ratios are proportional. **Reason (R):** The cross product of two collinear vectors is always a zero vector. * **Chapter:** Vector Algebra * **Probability Tag: High**

SECTION B: Very Short Answer Type Questions (2 Marks Each x 5 = 10 Marks)

This section tests formula application requiring 2-3 logical steps [3].

Q21. Evaluate $\tan(\sin^{-1}(\frac{3}{5}) + \cot^{-1}(\frac{3}{2}))$. [17] * **Chapter:** Inverse Trigonometric Functions * **Probability Tag: Very High**

Q22. Find the area of the region bounded by the curve $y^2 = 4x$ and the line $x = 3$. * **Chapter:** Application of Integrals * **Probability Tag: High**

Q23. If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$, prove that the vectors $\vec{a}$ and $\vec{b}$ are perpendicular to each other. [12] * **Chapter:** Vector Algebra * **Probability Tag: Very High**

Q24. Solve the differential equation: $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$. * **Chapter:** Differential Equations * **Probability Tag: High**

Q25. Find the probability distribution of the number of heads in two tosses of a fair coin. * **Chapter:** Probability * **Probability Tag: High**

SECTION C: Short Answer Type Questions (3 Marks Each x 6 = 18 Marks)

These are step-by-step derivation or procedural questions [7].

Q26. Show that the relation R in the set $A = \{x \in \mathbb{Z} : 0 \le x \le 12\}$, given by $R = \{(a,b) : |a-b| \text{ is a multiple of } 4\}$ is an equivalence relation. Find the set of all elements related to 1. [18] * **Chapter:** Relations and Functions * **Probability Tag: Very High**

Q27. Differentiate $x^x + (\sin x)^{\log x}$ with respect to x. [19] * **Chapter:** Continuity and Differentiability * **Probability Tag: Very High**

Q28. Evaluate using partial fractions: $\int \frac{x^2}{(x^2+1)(x^2+4)} dx$. [9] * **Chapter:** Integrals * **Probability Tag: High**

Q29. Find the area bounded by the curve $y = x^2$ and the line $y = x$ in the first quadrant. [11] * **Chapter:** Application of Integrals * **Probability Tag: High**

Q30. Solve the linear differential equation: $(1+x^2)\frac{dy}{dx} + 2xy = \frac{1}{1+x^2}$. [11] * **Chapter:** Differential Equations * **Probability Tag: High**

Q31. Evaluate the definite integral using standard properties: $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}} dx$. * **Chapter:** Integrals * **Probability Tag: Very High**

SECTION D: Long Answer Type Questions (5 Marks Each x 4 = 20 Marks)

This is the most critical section. It always features standard, calculation-heavy procedures [7, 20].

Q32. Using the matrix method, examine consistency and solve the following system of linear equations: $2x + 3y + 3z = 5$ $x - 2y + z = -4$ $3x - y - 2z = 3$ [8, 21] * **Chapter:** Matrices and Determinants * **Probability Tag: Very High**

Q33. Find the intervals in which the function $f(x) = 2x^3 - 3x^2 - 36x + 7$ is strictly increasing or strictly decreasing. Also, find the points of local maxima and local minima. [19, 22] * **Chapter:** Application of Derivatives * **Probability Tag: Very High**

Q34. Find the shortest distance between the lines $\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 4\hat{k})$ and $\vec{r} = (2\hat{i} + 4\hat{j} + 5\hat{k}) + \mu(3\hat{i} + 4\hat{j} + 5\hat{k})$. [23, 24] * **Chapter:** Three-Dimensional Geometry * **Probability Tag: Very High**

Q35. Solve the following Linear Programming Problem graphically: Maximise Profit $Z = 50x + 60y$ Subject to constraints: $5x + 8y \le 200$ (cutting limit) $10x + 8y \le 240$ (assembling limit) $x \ge 0, y \ge 0$. [25] * **Chapter:** Linear Programming * **Probability Tag: Very High**

SECTION E: Case-Study Based Questions (4 Marks Each x 3 = 12 Marks)

These assess real-world application using multi-part questions (1+1+2 marks) [3, 26].

Q36. Case Study 1: Determinants & Matrices [27] Three shopkeepers Ram, Shyam, and Ghansham are pushing for eco-friendly carrying bags. They use polythene bags, handmade bags, and newspaper envelopes. The numbers used by Ram, Shyam, and Ghansham are (20, 30, 40), (30, 40, 20), and (40, 20, 30) respectively. Their total spending on these bags is ₹250, ₹270, and ₹200 respectively. *(i)* Formulate the matrix equation representing this real-world problem. *(1 Mark) (ii)* Find the determinant of the coefficient matrix. *(1 Mark) (iii)* Find the exact cost of one handmade bag and one newspaper envelope using the matrix method or Cramer's rule. *(2 Marks)* * **Probability Tag: High**

Q37. Case Study 2: Application of Derivatives [28] A man has an expensive square-shaped piece of golden board of side 24 cm. He wishes to make an open box from it by cutting equal squares from the four corners and folding up the flaps. Let x be the side of the square cut from each corner. *(i)* Express the volume V of the open box as a function of x. *(1 Mark) (ii)* Find $\frac{dV}{dx}$ and equate it to zero to find critical points. *(1 Mark) (iii)* Use the second derivative test to find the value of x for which the volume is maximum, and calculate this maximum volume. *(2 Marks)* * **Probability Tag: Very High**

Q38. Case Study 3: Probability (Bayes' Theorem) [29] In an office, a company conducts a mandatory health checkup. A blood infection affects 5% of the population. The probability of a false positive on the test for this infection is 4%, while the probability of a false negative is 3%. An employee is selected at random and takes the test. *(i)* What is the probability that the person is actually infected and tests positive? *(1 Mark) (ii)* What is the total probability that the test comes back positive? *(1 Mark) (iii)* If a person tests positive for the infection, what is the exact probability that they are *actually* infected? *(2 Marks)* * **Probability Tag: Very High**

Expert Strategy Note for 2027: The heavyweights are Section D (20 marks) and Section E (12 marks). Solving the System of Linear Equations (Q32), Shortest Distance (Q34), Graphical LPP (Q35), and Bayes' Theorem (Q38) are essentially guaranteed templates [25, 30-32]. Secure these 18-20 marks first before diving deeply into complex integral calculus.

CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.